

PROGRESS IN DOCUMENTATION

DIGITAL LIBRARY RESEARCH: MAJOR ISSUES AND TRENDS

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Digital library research has attracted much attention in the most developed, and in a number of developing, countries. While many digital library research projects are funded by government agencies and national and international bodies, some are run by specific academic and research institutions and libraries, either individually or collaboratively. While some digital library projects, such as the ELINOR project in the UK, the first two phases of the eLib (Electronic Libraries) Programme in the UK, and the first phase of DLI (Digital Library Initiative) in the US, are now over, a number of other projects are currently under way in different parts of the world. Beginning with the definitions and characteristics of digital libraries, as proposed by various researchers, this paper provides brief accounts of some major digital library projects that are currently in progress, or are just completed, in different parts of the world. There follows a review of digital library research under sixteen major headings. Literature for this review has been identified through a search on *LISA* CD-ROM database, and a Dialog search on library and information science databases, and the resulting output has been supplemented by a scan of the various issues of *D-Lib Magazine* and *Ariadne*, and the websites of various organisations and institutions engaged in digital library research. The review indicates that we have learned a lot through digital library research within a short span of time. However, a number of issues are yet to be resolved. The paper ends with an indication of the research issues that need to be addressed and resolved in the near future in order to bring the digital library from the researcher's laboratory to the real life environment.

INTRODUCTION

Research in digital libraries is flourishing all over the world. Many of these research projects are funded by government agencies and national and international bodies, while others are run by specific academic and research institutions and libraries, either individually or collaboratively. In the United States, the six large projects under the First Digital Libraries Initiative (DLI-1) (http://www.cise.nsf.gov/iis/dli_home.html) supported by NSF, NASA and DARPA are in the

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process of completion. The Second Digital Libraries Initiative (DLI-2) in the US is about to begin, and this time the nature of support has broadened and the budget has doubled, and in addition to the three agencies in DLI-1, the sponsoring agencies in DLI-2 (<http://www.nsf.gov/pubs/1998/nsf9863/nsf9863.htm>) include the National Library of Medicine, the Library of Congress, and the National Endowment for the Humanities. The European Union is supporting projects through DELOS, a Digital Library Working Group (<http://www.iei.pi.cnr.it/DELOS/>) involving fifteen partners from various countries, as part of ERCIM – the European Research Consortium for Informatics and Mathematics (<http://www.ercim.org/activity/index.html>). In the UK, eLib, the Electronic Libraries Programme (<http://www.ukoln.ac.uk/services/elib/>) supports over sixty digital library projects.

In addition to the participating institutions in the digital library initiatives, a number of institutions now maintain web sites containing information on digital library resources, such as the BUBL website (<http://link.bubl.ac.uk:80/digitallibraries>), the UKOLN (UK Office for Library and Information Networking) website (<http://www.ukoln.ac.uk/metadata>), the British Library Research and Innovation Centre's website (<http://www.bl.uk/services/ric/diglib/digilib.html>), the *DLib Edu* website at Rutgers University (<http://www.scils.rutgers.edu/lis/digitallib.html>), the IFLA (<http://www.ifla.org/II/diglib.htm>) website, and so on.

A number of researchers have prepared bibliographies on different aspects of digital library research that can help us keep track of the huge volume of print as well as electronic information resources on digital libraries. Examples of such resources include the following:

- *Annotated bibliography of digital library related sources* compiled by Steven Ketchpel (<http://robotics.stanford.edu/users/ketchpel/annbib.html>)
- *Digital libraries: a selected resource guide* by Katharina Klemperer and Stephen Chapman (http://www.lita.org/ital/1603_Klemperer.htm)
- *Education-Line* (<http://www.leeds.ac.uk/educol/>)
- IBM Digital Library (<http://www.software.ibm.com/is/dig-lib/>)
- *Required Readings for Digital Libraries* LIS 2970 (http://www.sis.pitt.edu/~diglib/Required_readings_electronic.html)
- *Scholarly Electronic Publishing Bibliography*. Maintained by Charles W. Bailey, Jr. (<http://info.lib.uh.edu/sepb/sepb.html>), and so on.

There are also a number of journals that specifically deal with digital library issues, viz.:

- *Ariadne* (<http://www.ariadne.ac.uk>)
- *Biblio Tech Review Information Technology for Libraries* (http://wkweb3.cableinet.co.uk/biblio/html/may_98.html/)
- *D-Lib Magazine* (<http://www.dlib.org/>)
- *Initiatives in Digital Information* (<http://www.lib.umich.edu/libhome/IDINews/>)
- *International Journal on Digital Libraries* (<http://link.springer.de/link/service/journals/00799/index.htm>)
- *Journal of Digital Information* (<http://journals.ecs.soton.ac.uk/jodi/>)

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- *Journal of Electronic Publishing* (<http://www.press.umich.edu/jep/>)
- *Knowledge and Information Systems: an International Journal* (<http://kais.mines.edu/~kais/>)
- *The National Digital Library Program Reports, Library of Congress* (<http://icweb.loc.gov/ndl/per.html/>)
- *RLG DigiNews* (<http://www.rlg.org/preserv/diginews/>)

With the growing number of publications on different areas of digital library research, it has become very difficult for students, researchers and academics to keep track of the latest developments in this field. These publications, together with the popular press, have created a lot of hopes as well as myths about digital libraries. Some of the promises of digital libraries of the future, as well as some new challenges facing the digital library research community, have been discussed by Kung and Cleveland [1].

This paper briefly reviews the literature on different areas of digital library research. It is not claimed to be an exhaustive state-of-the-art review of the field. However, a large number of published articles, printed as well as electronic, and web resources of various institutions, research groups and individuals engaged in digital library research, have been studied to write this paper. This paper first aims to highlight the definitions and characteristics of digital libraries as proposed by various researchers. It then provides brief accounts of some major digital library research projects that are currently in progress, or are just completed, in different parts of the world. This will help build an idea of the nature of the mammoth research activities in this area, and also will help identify the major areas of research. The paper then reviews the literature sources under sixteen different headings.

THE DIGITAL LIBRARY: BASIC CONCEPTS

The terms 'digital library' and 'electronic library' are used interchangeably in the literature, though the former has become more popular over the years. There is another term, 'virtual library', that is also sometimes used interchangeably with digital library though, as Fecko [2] argues, they are not quite the same: a digital library consists of a networked collection of multimedia information typically available in one location, while a virtual library comprises a set of links to various information sources on the Internet. Several definitions of digital library are available in the literature (see for example, references 3–5, and <http://www.bl.uk>). While such definitions complement each other, there are some counter arguments too. For example, Miksa and Doty [6] raise the question why a digital library, or an electronic library, or a virtual library should be called a 'library'? They further suggest that a digital library might well be called a digital information system, or a digital publishing system.

For the purpose of this paper we shall take the following connotation of a digital library that was the outcome of a March 1994 Digital Library Workshop [7]:

1. A digital library is an assemblage of digital computing, storage and communications machinery together with the content and software needed to reproduce, emulate and extend the services provided by conventional libraries based on paper and other material means of collecting, cataloguing, finding and disseminating information.

2. A full service digital library must accomplish all the essential services of traditional libraries and also exploit the well-known advantages of digital storage, searching and communication.

The type of data involved in a digital library consists of texts, numerical data, figures, photographs, sound, video, films, slides, etc. In some sense a digital library encompasses all that is needed in terms of storage, processing and communications for applications such as multimedia, video servers and video conferencing [8]. The following are some of the major characteristics of a digital library [4, 7]:

- users are usually somewhere other than the information they want, and often wish to correlate things from several sources;
- users span the publishing/end-use spectrum, from authors and publishers to traditional library functions to end-seekers of information;
- for the foreseeable future a large percentage of important materials will be in non-digital form and the digital library needs to integrate meta-information about these as a major aspect of its services, rather than concentrating only on fully online materials;
- the digital library should support a wide range of service models for security, authentication, intellectual property control, billing and payment etc.;
- the focus of the infrastructure of digital libraries should be on facilitating the provision, dissemination and use of information by institutions and groups; materials in the digital library will be in a variety of media forms, including text, image, audio, video, 3-D models etc.;
- a significant part of the information in a digital library is the meta-information about items, such as annotations, catalogues, indexes etc.; and
- in the long term, the digital library will be based on platforms that are standard in the commercial world; a key issue in the overall design is a concern with compatibility and interoperation with other standards and integrative systems.

Technical issues to be addressed by digital libraries include distributed processing, gigabyte networks, optical storage, resource discovery, data mining, user interfaces and scaling in terms of size of databases, number of users, number of sites, geographical distribution etc., while the related issues deal with copyright, security, quality of presentation and availability [9]. However, a large part of the technical agenda is shaped by the need for robust methods of storage, organisation and dissemination of digital materials [10]. While technological advances – in storage, computing speed and networking – play a vital role, content-based searching and resource discovery, visualisation, knowledge discovery etc., are emerging as equally important research areas for serving user needs. There are other issues too. For example, Renear [11] argues that investigation of documents – of their essential nature and composition – must be at the core of the digital libraries' research agenda. In their introduction to the first issue of the *International Journal on Digital Libraries*, Adam and Yesha [12] mention that addressing the challenges in the area of DL requires a multidisciplinary approach

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that includes: acquiring and storing information, information providers, quality level of information, finding and filtering information, classification and indexing, query expansion and refinement, securing information and auditing access, and user interface.

These and many other socio-economic challenges in developing a digital library have been addressed by a number of digital library projects. Several universities are now offering courses on the digital library. Such courses address various fundamental as well as research issues. A look at the outline of the following courses would reveal this fact:

- the Rutgers University course on digital libraries addresses fundamental and integrating issues and problems that reflect efforts and thinking on digital libraries from a number of disciplines. Topics discussed include (<http://www.scils.rutgers.edu/lis/dlibresources.html>):
 - fundamental and theoretical aspects and questions underlying digital libraries;
 - nature of digital library collections;
 - organisation of digital libraries;
 - access to digital libraries;
 - use of digital libraries;
 - technical infrastructure;
 - social and economic issues;
 - integration of information resources;
 - relations to traditional libraries;
 - current digital libraries projects and initiatives.
- UKOLN, the UK Office for Library and Information Networking, based at the University of Bath, and the University of Bath Library have both been involved with teaching a course in the University of Bath MSc in Information and Library Management (<http://hosted.ukoln.ac.uk/msc/1998/>). This MSc course has a unit called 'Advanced information systems: towards the digital library' that aims to:
 - explore the concept of the digital library and the implications it will have on library services;
 - develop an understanding of the Internet and networked information;
 - develop suitable Internet skills;
 - identify issues that are preventing the development of the digital library;
 - gain an understanding of the future development of library services.

RESEARCH AREAS COVERED IN THIS PAPER

This paper aims to review those research publications, both in printed and electronic form, that have appeared in the recent past describing various research activities in different areas of digital libraries. However, as this is one of the fastest growing areas of research, and a number of publications are coming out every day in different forms and formats, it is very difficult to cover all of them.

Consequently this paper does not claim to be an exhaustive review of digital library research. Nevertheless, it covers a significant number of publications. Home pages of some major digital library initiatives, contents of the digital library journals like *D-Lib Magazine* and *Ariadne*, as well as the printed electronic papers on digital libraries identified by a literature search through *LISA* CD-ROM and Dialog library and information science databases, have been studied to cover all the different areas of digital library research as exhaustively as possible. The discussions cover the following areas of research in digital libraries:

1. collection development;
2. development methodology and design issues;
3. user interfaces;
4. information organisation: classification and indexing;
5. resource discovery: metadata;
6. access and file management;
7. user studies;
8. information retrieval;
9. legal issues;
10. social issues;
11. evaluation of digital information;
12. evaluation of digital libraries;
13. standards;
14. preservation;
15. implications for library managers;
16. future directions.

Attempts have been made to highlight major research activities in each of these areas. Finally, the paper looks to the future of digital library research.

SOME DIGITAL LIBRARY PROJECTS

This section provides a glimpse of some major digital library projects that have been recently finished, or are currently under way, in different parts of the world. The discussions here will provide a general idea about what all these research projects are about, what issues they are trying to address, and so on. However, before all these, some major digital library research projects are briefly mentioned.

The ELINOR Electronic Library Project

ELINOR (Electronic Library INformation Online Retrieval), the electronic library project at De Montfort University, Leicester, UK, developed the first electronic library to be built in a UK university [13]. The project was essentially carried out in two phases: the first phase was designed to create a practical pilot system and, in so doing, it addressed some of the issues involved; while the second part investigated the expansion of a small scale electronic library to a larger, distributed library system. The ELINOR project, concluded in 1996, demonstrated that [14, 15]:

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- libraries may conveniently digitise and supply small collections of high demand materials over a campus network in text and Tagged Image File Format (TIFF); and
- the barriers to the development and implementation of digital libraries are not technological but involve copyright, user issues and sustainability.

The eLib Programme

The UK Electronic Libraries Programme (eLib) grew out of the efforts to implement some of the recommendations of the Follett report [16]. Rusbridge [17], the Director of the eLib Programme, comments that an important precursor of eLib was the UK Higher Education Libraries Review (chaired by Professor Brian Follett) that was set up by the Higher Education Funding Councils, which fund universities in the UK, to look into the needs of libraries across the sector. The call for the first phase of the eLib Programme was issued in 1994, and that for the second in 1995. The two phases of the Programme cost £15 million over three years; and in this period JISC, the Joint Information Systems Committee (a joint committee of those same funding councils, and the funding agency for the eLib Programme) spent about £24 million on data services and probably £50 million on networks [17]. The eLib Programme did not, in itself, attempt to create the electronic library; instead, it aimed to address a number of important possibilities which had been identified in the Follett report [16], and slowly grew into a research and development programme addressing a number of issues related to digital libraries. Rusbridge [17] highlights the Programme areas of the first two phases of eLib and discusses the lessons that have been learned from this research.

The British Library's Digital Library Programme

The British Library's Research and Innovation Centre (<http://www.bl.uk/services/ric/>) is leading the Library's Digital Library Programme. This digital library will consist of a critical mass of digitally held documents (words, still images, moving images, sound and any combination of these) made available on demand, within an organised and ordered service framework, to a user anywhere in the world at any time.

The six big Digital Library Initiatives (DLI) in the US

The first Digital Library Initiative (DLI-1) in the US is funded by the National Science Foundation (NSF), the Advanced Research Projects Agency (ARPA), and the National Aeronautics and Space Administration (NASA). The initiatives' focus is to advance the means to collect, store and organise information in digital forms, and make it available for searching, retrieval and processing via communication networks – all in user-friendly ways. The six projects funded by the first DLI for the five year period (1994–1998) are (http://www.cise.nsf.gov/iis/dli_home.html):

1. the University of Michigan Digital Libraries Research Project (University of Michigan, Ann Arbor; grant: US\$ 4 million);

2. Building the Interspace: Digital Library Infrastructure for a University Engineering Community (University of Illinois, Urbana-Champaign; grant: US\$4 million);
3. the Environmental Electronic Library: a Prototype of a Scalable, Intelligent, Distributed Electronic Library (University of California – Berkeley; grant: US\$4 million);
4. Informedia: Integrated Speech, Image and Language Understanding for Creation and Exploration of Digital Video Libraries (Carnegie Mellon University; grant: US\$4.8 million);
5. the Stanford Integrated Digital Library Project (Stanford University; grant: US\$3.6 million);
6. the Alexandria Project: towards a Distributed Digital Library with Comprehensive Services for Images and Spatially Referenced Information (University of California – Santa Barbara; grant: US\$4 million).

Library of Congress: National Digital Library (LC DL) Program

The National Digital Library Program at the Library of Congress is a five-year programme to assemble an initial core of American historical and cultural primary source material in digital form, selected for conversion from the Library's vast holdings of print and non-print materials [18]. One aspect of the National Digital Library Program (NDLP) involves selecting and digitising a heterogeneous array of resources, many of which are fragile. The NDLP also aims to provide convenient and effective access to the digital archive it assembles, access for a broad constituency of users, from scholarly researchers and professionals to college students and school children [18].

The Intelligent Digital Library Project

Liddy et al. [19] describe the Intelligent Digital Library Project that is being carried out by the Syracuse University's School of Information Studies (IST) in collaboration with a number of partners drawn from research and commercial information organisations, such as AskERIC, Scholastic Inc., Northeast Parallel Architecture, Center (NPAC), NYSERNet, Bell Atlantic, Discovery Communications, CNN Newsroom, NYNEX, Dialog Information Systems and Xerox Palo Alto Research Center (PARC). The major areas covered in this project include: intelligent retrieval via a natural language text retrieval system (DR-LINK), scale-up of intelligent retrieval in an HPCC environment, evaluation of digital libraries' impact on users' behaviour and performance, and policy implications of the creation, use and availability of a digital library [19].

Global-Info: a digital library project in Germany

Global-Info is a DL programme in Germany funded by the Federal Ministry for Education, Science, Research and Technology (BMBF). Three types of projects have been identified and expenditure of sixty million DM has been planned for a period of six years [20]. Three types of projects are supported: pilot projects to

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fund the basic needs for collaboration (including the development of further projects); an administration project to organise Global-Info; and the main research and development projects to be funded by Global-Info.

Australian digital library initiatives

Australian libraries (at the Federal, State and University levels), together with commercial and research organisations, are supporting a diverse set of digital library projects [21] that take on many technical and related issues. The major projects include: WORLD 1, Electronic Reserve Project, Electronic Journal Project; REDD: an Electronic Document Delivery Project; Hume Image Collection Project; DIGLIB: Queensland Country Towns Image Project; NSW Parliaments Newspaper Clippings and Press Releases Imaging Project; Preserving Oral History Recordings Project; Australian Cooperative Digitisation Project 1840–45; The Information Storage & Retrieval Project; Scholarly Electronic Text and Image Service; UNILINC Projects, LISWEB Project; The Documentary Images Project; Indian Ocean Rim Region Virtual Library Project; The Resource Discovery Project; Multimedia Source Project; and Australian Museums On Line.

New Zealand Digital Library Project

The New Zealand Digital Library Project is a research programme in the Computer Science Department at Waikato University [22]. This project aims to develop the underlying technology for digital libraries and make it available publicly so that others can use it to create their own collections. The project is concerned with large collections of electronic, predominantly textual, documents, physically dispersed on computers the world over, which are accessible through a uniform interface that allows information to be located and accessed. The NZDL Project (<http://www.cs.waikato.ac.nz/~nzdl>), illustrated by a large example collection of Computer Science Technical Reports, has shown that the new realities of Internet publishing are that information is provided in a widely distributed manner, and it is up to the consumer to locate what is needed [22]. Witten et al. [22] report that the New Zealand Digital Library Project is developing a novel software infrastructure that enables those who manage and maintain such collections to make them publicly available.

Canadian Initiative on Digital Libraries

The Canadian Initiative on Digital Libraries (CIDL) (<http://www.nlc-bnc.ca/cidl/cidlproe.htm>) is an alliance of Canadian libraries with the following objectives:

- to formulate and implement strategies towards increased communication and education on digital library matters;
- to promulgate digital library standards and best practices;
- to define methods to improve co-ordination of activities and avoid duplication in the development of digital resources;
- to establish strong relations with others in the information arena: creators, publishers, information technology industries, archives, museums, cultural agencies, government agencies at all levels, and individual and institutional users; and

- to raise awareness of Canadian digital library activities, both within and outside Canada.

The Digital Library Project in Japan

In Japan the digital library effort includes the preparation of a national union catalogue and also a set of experiments on primary material at the National Diet Library [23]. Another digital library initiative in Japan is the NACSIS project that has the following features (<http://www.nacsis.ac.jp/dl/dl-e.html>):

- online document database of Japanese academic societies;
- combines bibliographic databases and electronic document delivery system;
- users can get access to their target documents via the new 'just browsing' method as well as conventional Boolean search, search window, bibliographic data, document page, sample window;
- operates on high-speed ATM Network.

There are a number of other digital library research projects currently under way in different parts of the world. Some of these projects are:

DELOS: Digital Library Working Group (<http://www.iei.pi.cnr.it/DELOS/ErcimDL/>)

Its objective is to promote research into the further development of digital library technologies, in particular to stimulate research activities in areas which are relevant for the efficient and cost-effective development of digital library systems.

The Digital Library Federation (DLF) (<http://www.clir.org/digilib/dlhomepage.htm>)

It was founded in 1995 to establish the conditions for creating, maintaining, expanding and preserving a distributed collection of digital materials accessible to scholars, students and a wider public. The Federation is a leadership organisation operating under the umbrella of the Council on Library and Information Resources. It is composed of participants who manage and operate digital libraries.

ERCIM: the European Research Consortium for Informatics and Mathematics

(<http://www.ercim.org/activity/index.html>) It aims to foster collaborative work within the European research community and to increase co-operation with European industry. Leading research establishments from fifteen European countries are members of ERCIM.

IEEE Technical Committee on Digital Libraries (http://cimic.rutgers.edu/ieee_dltf.html)

In 1995, the IEEE Computer Society established the Task Force on Digital Libraries as a first step leading to a full-fledged Technical Committee. The task force is to promote research in the theory and practice of all aspects of digital libraries.

NDLTD: The Networked Digital Library of Theses and Dissertations (<http://www.ndltd.org>)

is a digital library research project at Virginia Tech, USA, that maintains a large digital library of theses and dissertations, and allows students to develop and submit theses and dissertations electronically.

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REVIEW OF DIGITAL LIBRARY RESEARCH

A discussion of publications pertaining to specific areas of research on the digital library is presented under sixteen different headings, where fifteen headings include various digital library research areas and the last one discusses future directions.

Collection development

This is one of the most important stages, and often is a major stumbling block, in digital library development. However, several measures have been taken by researchers to develop effective collection development policies for digital libraries, and consequently a number of publications have appeared in this area. Some of these just describe the collection development policies adopted in a particular digital library project, while others address policies and issues related to digital collection development in general. Weiner [24] discusses the future of the e-journal considering such issues as pricing, implementation, distribution, copyright and format from the perspectives of publishers, librarians and editors. Galbreath [25] discusses the role of collection development in libraries that are rapidly changing from conventional libraries to digital libraries where most material will be available electronically. Kluiters [26] reports on the first conference on Elsevier Electronic Subscriptions (EES) (held on Oct 17–18, 1996 at Château Marquette in Heemskerk, the Netherlands), that contains the contributions of the first 'live' experiences on organisational, operational and technological levels in building digital journal collections at participating libraries in a commercial environment.

The National Library of Australia's PANDORA project [27] uses particular selection guidelines developed by the Selection Committee on Online Australian Publications (SCOAP). This project has found out that developing and applying collection management policies may be time-consuming and expensive in the short term but attempting to collect all digital documents will not be sustainable (or even desirable) in the long term. Witten and McNab [28] discuss the underlying technology used in the New Zealand Digital Library Project for creation and use of digital library collections comprising 40,000 technical reports on computer science collected from over 300 sites Worldwide and Computerists' Communique; FAQ Archive; HCL Bibliography; Indigenous Peoples; Oxford Text Archive; Project Gutenberg Collection; and TidBITS. They also discuss policies on collection development relating to the new collections including music and audio collections.

Butler [29] suggests that disintermediation resolves the difficulty in obtaining use rights from publishers, and distributed computing enables the library to convert part of its collection to digital form to make materials more usable. Both these trends are essential to the development of course reserves which are a prototype application for digital library collections linked through virtual library networking. Enderle [30] describes the collection development policies adopted by German digital library projects with particular reference to the development of shared digital holdings; the development of information servers; access to shared digital holdings; the boundary between free access to subscribers and commercial publishing products; and standardisation of the document format.

The Alexandria Digital Library (ADL) comprises a set of Internet nodes implementing combinations of the four primary ADL architecture components –

collections, catalogues, interfaces and ingest facilities (which a digital library uses to add documents and information about document cataloguing and access) [31]. The California Digital Library project (CDL), announced in October 1997 by the University of California, came up with a set of general guidelines that may be used for building the digital library collection in any university [32].

NCSTRL (<http://www.ncstrl.org>), a federation of over 100 institutions with the goal of providing a federated library of computer science materials, has developed and adopted an open architecture of digital libraries. This means that the functionality of the digital library is partitioned into a set of well-defined services or functions, each with a well-defined protocol specifying the interface to that service [33]. This allows the participating organisations to develop or use different designs and implementations of the services, as long as their interfaces are consistent with the agreed protocol. The open architecture supports the federation concept in two ways:

- it provides a modular construct for each organisation to create its own digital library system, through the selection of the best technology products to serve the needs of the target users; and
- the architecture provides a means for defining the interfaces between the digital libraries of the federation that is achieved through specification of interfaces to the services by each separate library.

Development methodology and design issues

The nature and characteristics of DL systems call for mechanisms for the dynamic adaptation of the system to the constantly changing operational environment [34]. In the development of digital library systems various models have been proposed. For example, the University of Stanford's view of DL is a shared information bus that connects various information sources. On the other hand, the University of Michigan's view is of a collection of collaborating agents. The common denominator in all these different views is that a DL system consists of a number of servers, spread over the Internet, that interact with each other in order to meet user requests.

Zhao [35] describes the PDL (Personal Digital Library) designed by the ELINOR Project team. PDL is a personal information system that acts as a front end to other electronic library systems and manages all the users' information needs. Several other design methodologies have been proposed and tested by researchers. For example, Kipp, Newcomb and Newcomb [36] describe a new model called HyTime Document. FEDORA is a digital object and repository architecture, developed at Cornell University, USA, for digital libraries that is positioned within a large open-architecture framework in which the total functionality of a digital library is partitioned into a set of services with well-defined interfaces. The core services include the following [37]:

1. repository services that provide the mechanisms for depositing, storing and accessing digital objects;
2. index services that provide the mechanisms for discovering digital objects;

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3. collection services that provide the means of aggregating sets of digital objects and services into meaningful collections;
4. naming services that register and resolve globally unique, persistent names for digital objects; and
5. user interface services that provide a human gateway into the other services.

Since 1994, the Alexandria Digital Library (ADL) has developed three architectures for digital libraries of georeferenced information at UCSB (University of California at Santa Barbara, USA) [38]:

1. the first or the 'rapid prototype' architecture employed a desktop geographic information system (GIS) as the user interface to a single ADL catalogue database;
2. the second, or the 'Web prototype' architecture replaced the GIS with an HTTP server, which presented a user interface of dynamically-generated HTML pages;
3. the most recent of these, the third architecture implemented in July 1998, is based on three-tier client-server architecture that relies heavily on a middleware layer to present a single uniform set of interfaces to multiple heterogeneous servers (generalisations of library catalogues). These standard interfaces, all of which are implemented in HTTP, support session management, collection discovery and evaluation, metadata searching, metadata retrieval and online holding retrieval. These interfaces are intended to be generic enough to support arbitrary clients, of which the ADL client is the first sample implementation.

Schwarzwalder [39, 40] explains how digital journals can be added to existing services and suggests models for partnering with information publishers that are appropriate to this new medium. In the third in a three-part series of articles dealing with building the science and technology digital library, Schwarzwalder [41] discusses options for adding services to World Wide Web libraries and integrating them with an existing information organisation. He further explains how a digital library can integrate information and services, and manifest the same information ergonomics that can be seen at the library information desk. Collier [42] presents the lessons learned about electronic libraries from the range of research projects carried out at De Montfort University since 1991. He discusses the setting up of the Institute for Electronic Library Research and the current pilot research project, to set up an electronic library based on IBM's Digital Library Architecture (DLA) with an IBM SP2 supercomputer as the hardware platform.

User interfaces

User interface is an area which has attracted much attention from information retrieval researchers and one can see significant improvements in the interface design on online and CD-ROM databases available today compared to those of, say, a decade ago [43]. The issue of the user interface becomes very critical in a digital library environment, because information may come from a variety of different sources, in different forms and formats, and virtually anyone on earth can

be a user. Consequently, several attempts have been made to improve user interfaces. Spink et al. [44] have undertaken a research project to investigate the nature, manifestations, and behaviour of successive searching by users in digital environments, and to derive criteria for use in the design of information retrieval interfaces and systems supporting successive searching behaviour. Edelson and Gordin [45] argue that the value of digital libraries lies in the authentic activities that they can allow learners to engage in. They suggest that digital libraries need to be made accessible for learners through a variety of bridging strategies, such as: supportive interfaces, activities design, resource selection and organisation, and documentation. Cousins [46] describes the detailed design of an interface to a heterogeneous digital library at Stanford University. Considering the fact that a library's primary goal is to support user tasks, and since user tasks involve an increasingly rich variety of services, the user interface designed at Stanford University aims to achieve the following goals [47]:

- to integrate the results from a broad array of services;
- to design the interface to handle widely varying time scales;
- to let the user know before initiating a service whether it will take milliseconds or hours to complete;
- to provide feedback on the progress of the service, and a means for interrupting the service;
- if the user has moved on to another task, the interface should continue to accept results from running services and compile them into a meaningful form for when the user returns to this task;
- to make the system extensible; and
- to support sharing and reuse of information-processing knowledge.

Some user interfaces need to have some special characteristics because of the data that they handle. For example, the user interface of the ADL (at the University of California at Santa Barbara) has the following features because it deals with spatial data (<http://www.alexandria.ucsb.edu/>):

1. a 'zoomable' background map that provides a basis for spatial queries;
2. a query form that accepts a thematic keyword, the type of dataset, bounding co-ordinates, and time;
3. a table showing summary metadata about items in the collection that satisfy specific criteria;
4. a window showing detailed metadata about a specific item, which is displayed to the user as soon as an item is selected from the previous table;
5. a set of thumbnail sketches or icons of items held in the collection;
6. a set of footprints displayed on the base map, showing the search area and the items found;
7. displays of the specific raster items held in the RPS store;
8. displays of specific vector items held in the RPS store;
9. displays of video and remote web sites;
10. hypertext document retrieval using a web browser, including hyperlinks to other objects in the library.

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Information organisation: classification and indexing

Libraries and information centres have been engaged in organising information for a long time, and the beginning of modern classification and cataloguing can be traced back to 1876 with the appearance of the first edition of the Dewey Decimal Classification (DDC) and Cutter's *Rules for a dictionary catalog*. Since then a huge amount of research effort has gone into this field resulting in tremendous improvements in the principles and practices of classification, cataloguing, and in the process of bibliographic organisation and control at large. The recent developments in the Internet and WWW have brought new challenges to information professionals and consequently a number of researchers are now engaged in organising information on the web [43]. Dolin et al. [47] argued that as the number and variety of sources increase, new ways of automatically summarising, discovering and selecting collections relevant to a user's query are needed. One such method involves the use of classification schemes, such as the Library of Congress Classification (LC), within which a collection may be represented based on its content, irrespective of the structure of the actual data or documents. For such a system to be useful in a digital library environment, it must be easy to use for both collection managers and users. It must be possible to classify documents automatically within a classification scheme, and there must be a straightforward and intuitive interface with which the user may use the scheme to assist in information retrieval.

The Internet Public Library (IPL), a WWW based service provided by the University of Michigan School of Information, makes available a wide range of full-text files arranged according to DDC [48]. The ADL project team have designed and tested Pharos (<http://pharos.alexandria.ucsb.edu/>), an architecture that accommodates heterogeneity in content and format, among multiple sources as well as within a single source including web sites, FTP archives, newsgroups, and full digital libraries. Pharos is based on the use of hierarchical classification schemes. These include not only well-known 'subject' (or 'concept') based schemes such as DDC and LC, but also, for example, geographic classifications, which might be constructed as layers of smaller and smaller hierarchical longitude/latitude boxes.

CyberStacks (<http://www.public.iastate.edu/~CYBERSTACKS/>) is a prototype created to facilitate access to selected Internet and World Wide Web (WWW) resources in science and technology through a comprehensive application of LC. It was initiated as a project to demonstrate and explore the use of such a scheme to facilitate access to selected WWW resources by current staff and potential clientele of the Science and Technology Section of the Reference and Instructional Services of the Iowa State University Library, Ames, Iowa.

SCORPION (<http://purl.oclc.org/scorpion>) is a research project at OCLC that explores the indexing and cataloguing of electronic resources. The primary focus of SCORPION is building tools for automatic subject recognition based on well-known library classification schemes like DDC that is maintained electronically via the Editorial Support System (ESS) at OCLC Forest Press. The corresponding ESS records contain all the raw information used to produce the printed DDC Schedules and Tables. ESS records include a variety of labelled fields. Some or all of these fields can be used to build ranked retrieval databases for automatically

assigning subject codes to documents. By treating each document as a query against such a database, the result set can be viewed as possible subjects for the document. SCORPION uses a clustering technique based on the measures of similarity between two documents. These similarity measures are calculated as a function of the frequency of words in the two documents. SCORPION measures the similarity between an input document and the concept documents as represented by the ESS records. Clustering measures the similarity between every pair of documents in a given collection. Subramanian and Shafer [49] suggested that if SCORPION were run against a complete document collection, then the results could be used to partition the collection in a manner similar to clustering.

Another example of the use of DDC as a browser can be seen in the HOPAC (Hierarchical OPAC) interface reported by Allen [50]. This interface uses DDC hierarchy to facilitate browsing and searching. Allen reports that the HOPAC system has been implemented and runs with over 50,000 book records, and thus demonstrates that the structure of a classification system can be used as an aid for searching and navigating the digital library catalogue.

Banerjee and Mittal [51] argue that the specific characteristics of DL will require that an efficient indexing scheme allowing easy retrieval across a distributed representation be developed. Consequently, they have reviewed some of the previous approaches and then propose a new approach, based on the use of a large and sophisticated linguistic ontology, to generate suitable keyword matches. Their approach makes use of a large linguistic ontology – the WordNet lexical database, a very large lexical database consisting of approximately 100,000 different word forms, organised into 70,000 sets of synonyms – to generate alternative search patterns.

Vizine-Goetz [52] describes the *Subject headings for children (SHC)*, a simple-to-use list of Library of Congress juvenile subject headings supplemented with suggested Dewey numbers. Published by OCLC Forest Press in 1994, *SHC* is intended to assist cataloguers and reference users, including parents and children, in identifying subject headings assigned to children's works and the Dewey numbers commonly associated with them. *SHC* is produced as a collaboration between the OCLC Office of Research, the Dewey editorial team, and the work's editor, Lois Winkel. In this project, computer processing of WorldCat records is combined with in-depth human review to produce juvenile heading/DDC number pairings [52].

Shapiro and Yan [53] report the Electronic Libraries and Information Highways MITRE sponsored research project that aims to help researchers working in digital libraries increase their chance of matching the language of authors. The project aimed to determine whether query formulation can be improved through the addition of semantic knowledge that is interactively gathered from a thesaurus that exists in a distributed, interoperating, co-operative environment.

There are several other attempts at organising digital information resources using traditional classification and indexing tools: CyberDewey (<http://ivory.lm.com/~mundie/DDHL/cyberDewey.html>) a site employing the Dewey Decimal Classification (DDC) developed by David A. Mundie; EELS, the Electronic Engineering Library, Sweden (<http://www.ub2.lu.se/eel/eelhome.html>); and DutchESS, the Dutch Electronic Subject Service (<http://www.konbib.nl/dutchess>). The Internet Scout Project (<http://scout.cs.wisc.edu/scout/index.html>)

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is one of the outstanding projects that has applied a traditional controlled vocabulary (Library of Congress Subject Headings (LCSH)) to enhance access to a select collection of web resources.

Resource discovery: metadata

The separation between metadata and data is extremely useful, particularly in a DL environment, as it allows us to query not only the currently stored information but also the structure of information [54]. Dempsey and Heery [55] define metadata as data associated with objects which relieve their potential users of having to have full advance knowledge of the existence of characteristics. Younger [56] defines metadata as documentation about documents and objects; they describe resources, indicate where they are located, and outline what is required in order to use them successfully. Several metadata schemes have been created by library and information professionals such as the MARC formats, the AACR2 catalogue formats, subject headings lists (such as the LCSH), and classification schemes such as LC, DDC and so on. Each such scheme is constructed by experts in the field with knowledge of the specific domains, information resource needs and the requirements for describing documents.

Adam et al. [54] suggest that in a DL setting, one expects a variety of data types and within these data types a number of sources of information. There will thus be a great need for just finding out what information is contained in a DL. This means that the concept of metadata will need to be extended by information models that present meta-information in concise and meaningful ways, and from which the user may decide what to retrieve [54]. While various metadata schemes have been used for bibliographic access and control for decades, there remains a question of how to catalogue and index all those materials available on the Internet using these schemes. This has given rise to a thought that electronic documents need to be *self-indexed* (as opposed to the assignment of cataloguing and indexing tags, and value, added by cataloguing and indexing agencies). However, it is obvious that in order for the documents to be self-indexed, a core set of metadata elements needs to be identified and each creator of electronic documents should be able to implement it in the record that he or she creates. With this objective, a simple resource description set of data has emerged which is known as the Dublin Core (http://purl.org/metadata/dublin_core).

The Dublin Core is a metadata element set intended to facilitate discovery of electronic resources. The Dublin Core Workshop Series has gathered experts from the library world, the networking and digital library research communities, and a variety of content specialities in a series of invitational workshops. The building of an interdisciplinary, international consensus around a core element set is the central feature of the Dublin Core. The Dublin Core metadata set prescribes fifteen elements: (<http://www.lub.lu.se/cgi-bin/nmdc.pl>), title, author or creator, subject or keywords, description, publisher, other contributors, date, resource type, format, resource identifier, source, language, relation, coverage, and rights management.

Weibel [57, 58] suggests that in order to enable information creators to apply the metadata, a mechanism for embedding the data within HTML documents has to be established. The World Wide Web Consortium (W3C) (<http://www.w3.org>) has begun implementing an architecture for metadata for the web. The Resource

Description Framework, or RDF (<http://www.w3.org/Press/RDF>), is designed to support the many different metadata needs of vendors and information providers. Representatives of the Dublin Core effort are actively involved in the development of this architecture, bringing the digital library perspective to bear on this important component of the web infrastructure. In 1996, there was a conference, organised by UKOLN and OCLC, to examine the various metadata issues including the Dublin Core. This meeting gave rise to a proposal, named after the place of the conference, the Warwick Framework proposal [59, 60] calling for an architecture for the interchange of various metadata packages.

In order to meet some specialised requirements, Dublin Core can be customised with additional elements or qualifiers, though such refinements can compromise interoperability across applications. The problem of interoperability is compounded when we consider Dublin Core in translation. Baker [61] reports that so far we have freely invited metadata-using institutions in many countries to translate Dublin Core into their own languages, and all such efforts spread awareness of the standard, thereby increasing the number of potential users and the value of Dublin Core for cross-language interoperability. However, regional linguistic differences or institutional rivalries could eventually result in the creation of more than one Dublin Core for any given language. To resolve such problems, a peer review process may be introduced that may help maintain distinction between pure Dublin Core and local extensions or adaptations. Baker [61] recommends that through the mechanism of a distributed registry system, such as the one created at the Asian Institute of Technology, Bangkok, in twelve languages, certified Dublin Core elements in various languages could be offered for automatic loading into browser templates or metadata editors.

Lagoze [60] argues that recognising the limitations of the current search engines, researchers are now actively pursuing both standards for descriptive surrogates for networked objects and methods for associating surrogates with those objects. Such studies aim at developing a number of surrogate templates that would facilitate the resource discovery process. Chowdhury [62] proposes that the template mining approach can be used effectively for the resource discovery process; the metadata tags, such as subject and description, in the Dublin Core metadata format, for example, can be extended, and be made more structured, by specifying various templates.

Access and file management

Managing access to online information is a broad problem that occurs in a wide range of different applications. Access management has been a topic of digital library research for several years. Kahn and Wilensky [63] report a work that began as the Computer Science Technical Reports project. An updated version of the concepts derived from that project is included in a recent paper from the Cross-Industry Working Team [64]. Another overview of the field appears in the report of the National Science Foundation workshop in March 1996 [65]. Managers of online information wish to implement policies about who can access the information, under what terms and conditions. Arms [66] describes a general approach to this problem and experience in applying it to digital libraries.

The ATHENS (<http://www.athens.ac.uk>) access management system was introduced in 1996 and then extended in 1997 to provide access control facilities for all the national data centres and services in the UK. ATHENS has produced many benefits and has highlighted significant issues as reported by Wiseman [67]. Bowman and Camargo [68] argue that if we view the problem of sharing files among organisations as a problem of building and managing objects in a digital library, natural extensions to the file system interface that solve problems of collaboration become apparent. Services for explicit types, metadata management, information retrieval, and construction of meta-objects facilitate dramatic improvements in collaboration. They suggest that integration of file system and digital library technology reduces the time to develop collaborative applications and significantly improves integration of information from multiple repositories.

User studies

As digital libraries become a major source of information for many people, we need to know more about how people seek and retrieve information in digital environments. Library and information professionals have continuously engaged themselves in assessing user requirements and taking necessary measures to tune information retrieval systems to suit the changing environment; and this is also continuing in the Internet and web environment [43]. User studies also occupy a major area in DL research. Davies [69] describes the user issues surrounding the current and likely future developments in academic electronic libraries as successors to the ELINOR Project. Spink et al. [44] argue that users with a problem-at-hand and associated question-in-mind repeatedly search a literature for answers, and seek information in stages over extended periods from a variety of digital information resources. House et al. [70, 71] report the application of user needs assessment and evaluation in the iterative, user-centred design of a component of Cypress (<http://elib.cs.berkeley.edu/cypress>), a DL project at the University of California at Berkeley. This report is illustrative of the application of user-centred iterative design to digital libraries in general. It demonstrates how a relatively straightforward, moderate level of effort involving users, not just for feedback on an interface, but for an understanding of how they do their work, resulted in a significant improvement in the design of a component of the UC Berkeley project. User needs assessment study of the UC Berkeley DLI had the following goals [71]:

- to improve the understanding of the use of information for complex cognitive work;
- to develop conceptual bases and methods for evaluating digital libraries;
- to support the development of the UC Berkeley Digital Environmental Library by providing the other components of the project with an understanding of the users, their context and their work;
- to advise on content, functionality and interface design; and
- to provide feedback on designs and prototypes by means of heuristic evaluation, small-scale user feedback and larger-scale user testing.

A variety of qualitative methods have been used in the UC project including extensive interviews, learning about people's present work processes and tools

and their perceived needs for a digital library. Experiments have been done with a variation on thinking-aloud protocols that consists of videotaping users interacting with the system and then asking them to review and comment on their tapes, so the thinking-aloud process is retrospective. It has been argued that this minimises the disruption of the users' work while giving us the benefit of their sense-making process [71].

Spink et al. [44] report two related projects on user studies in a DL environment. The first project is based in the School of Library and Information Sciences at the University of North Texas and is funded by the National Science Foundation. The second project is based at the Department of Information Studies at the University of Sheffield (UK) and is funded by a grant from the British Library Research and Innovation Centre. Three questionnaires were used to collect data: reference, client post-search and searcher post-search questionnaires. Each successive search episode with a search intermediary for textual materials on the Dialog Information Service was audiotaped and search transaction logs were recorded. Quantitative analysis included statistical analysis using Likert scale data from the questionnaires and log-linear analysis of sequential data. Qualitative methods included: content analysis; structuring taxonomies; and diagrams to describe shifts and transitions within and between each search episode. Outcomes of the study are the development of appropriate model(s) for IR interactions in successive search episodes and the derivation of a set of design criteria for interfaces and systems supporting successive searching [44].

Information retrieval

Information retrieval has been a major area of research in the information field and the recent appearance of the Internet and WWW have proliferated IR research. Chowdhury [43] has reviewed the recent literature on IR research in the context of the Internet and WWW and has pointed out the following trends:

- researchers are trying to come up with a common user interface and a metadata set to let users know what is available; they are also trying to let the end-users easily access and retrieve the information irrespective of the software and hardware variations;
- traditional approaches to access to and use of remote information are changing fast, and traditional online information service providers are quickly adapting to the web technology;
- user studies has always remained a critical area and the user-oriented approach to design, development and evaluation of information systems is becoming more and more important in the digital library environment;
- the subject approach to information, which has traditionally been controlled by library and information professionals with the help of classification, cataloguing and vocabulary control tools, has become an important area of research in the Internet environment;
- intelligent search agents are being designed to perform a number of operations in the digital information environment; and

- in order to solve the existing problems, information professionals with their rich experience in organising and searching information, as well as in identifying user needs, should work hand in hand with software professionals in organising the so-called 'anarchy' of the Internet.

Bates [72] argues that many people in our society, including many in the Internet and digital library environment, assume that subject access to digital resources is no longer a problem, and it will be solved with a few small modifications of current full-text indexing systems. However, this is not the case, and in fact there are quite a few issues that need special consideration, which she mentions as follows:

- the human domain and user factors would still operate in a digital library;
- information retrieval systems involve language and cognitive processing and it will be difficult to automate them fully;
- information retrieval systems pose serious scalability problems; and
- user needs vary according to time, subject domain and a number of other factors, and therefore optimal indexing and retrieval mechanisms would substantially vary from one subject domain to another.

Lesk [73] describes the seven stages of development of IR studies and research, and forecasts that we are going to have a large collection of online books, though the publishing industry has to sort out the economics, and when we do have video libraries, we will need to build better retrieval systems through rigorous image and sound recognition research, and we will need to build new techniques in co-operative browsing. Bates [72] suggests that indexing and information system design need to proceed within the context of a deep understanding of the character of the intellectual domain, its culture and its research questions.

Twidale et al. [74] describe a research project that was designed to gain a better understanding of the nature of collaboration in the information-searching process, and to consider how information system design could better address and support collaboration activities. Kirriemuir et al. [75] describe subject gateways, that are also known as subject-based information gateways, subject-based gateways, subject index gateways, virtual libraries, clearing houses, subject trees, pathfinders and other variations thereof. They describe the characteristics of some of the subject gateways currently accessible through the web, and compared them with search engines, such as AltaVista. Carbonell [76] looks at various possibilities for digital library functions, such as interactive search formulation, automated analysis of user needs, sub-document search and indexing, sophisticated information retrieval, on-demand document summarisation, multi-document summarisation, multimedia search and indexing, access to live or near live information, active information sources, symbiotic human machine gurus, information on tap, and the universal library.

Arguing that the growth of interest in the electronic library concept and increasing reliance on multimedia materials have increased the demand for image databases as functional parts of digital libraries, Jain [77] discusses the requirements for these emerging information systems and presents query methods and

data models suitable for such systems. Visualisation techniques are now being tested by a number of IR researchers. The Big Picture (<http://www.public.iastate.edu/~CYBERSTACKS/BigPic.htm>) is a clearinghouse of activities that have applied information visualisation technologies to enhance use and access to web and selected non-web databases. Other notable examples of this approach include the work of Xia Lin, Drexel University (<http://lislin.gws.uky.edu/sitemap/sitemap.html>) and Hsinchun Chen of the University of Arizona (<http://ai.bpa.arizona.edu/et/>), a co-investigator with Bruce R. Schatz in the University of Illinois at Urbana-Champaign.

A number of researchers also see the digital library as a potential area for the application of knowledge-based information retrieval systems. For example, Lancaster [78, 79] discusses the ability of artificial intelligence and expert system technologies to contribute to some important functions of a digital library. Koller and Shoham [80] discuss the role of artificial intelligence in digital libraries, and report on the challenges presented to artificial intelligence by information agents acting as part of a digital library. They suggest that contributions from probability theory, knowledge representation, reasoning about uncertainty, and multi-agent systems can play a part in meeting these challenges. Walker [81] describes the Internet Knowledge Manager (IKM) which is an environment that provides an understandable way of representing knowledge as readable agents that can also be run as a software. With the IKM, one can use a standard web browser to write and run one's own agents. Such agents, written in simple English, can be placed in a digital library and selectively retrieved using text search techniques [81].

Legal issues

A number of legal issues are associated with digital libraries, and consequently a number of researchers are involved in this area. Research addressing the major legal issues such as copyright, charging and authenticity, are discussed in the following sections.

Copyright Valauskas [82] argues that libraries are handicapped in their move to digital collections by enormous legal problems in securing rights to the current literature, and solutions to this dilemma will include changes in the current copyright law and technological arrangements to protect the interests of the owners of intellectual property. Cornish [83] argues that though information is increasingly becoming an international commodity, copyright and associated intellectual property rights tend to be ruled by national legislation. It is therefore particularly important that the concerned parties come together to agree how to manage rights in an international electronic environment. Oppenheim [84] analyses the current position in the field of copyright, gives the basis for copyright legislation and discusses current reactions to the challenges posed by digital media. Gadd [85] presents a practical guide to gaining copyright clearance for making electronic copies of journal articles based on experience gained on the eLib project called ACORN. He also provides some tips on identifying and contacting copyright owners, elements to include in letters of approach, chase

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tactics, and dealing with refusals and charges. Okerson [86] suggests that if access to electronic materials without payment for every use is to be recognised, then fair use is the area in which the bridges can be built between the rights of copyright owners and those of information users. Demas and Brogdon [87] report the results of a copyright investigation as part of the project conducted at the Mann Library, Cornell University, New York, to determine the copyright status of brittle monographs to be scanned as part of the national preservation plan for agricultural sciences literature.

It is now commonly agreed that there is an urgent need to educate people about the importance of copyright. However, a number of questions may arise if and when we want to apply current copyright law to new digital formats and media, such as [86]:

- to what extent are the works on the newer electronic media protected by law?
- many works are being created through extensive electronic communities or collaborations – who owns and benefits from these?
- how can we track who owns what, assuming that ownership makes sense at all?
- how do we efficiently compensate information owners when their works can be sold by the word, phrase or even musical note?
- what are the liabilities of Internet access providers, who may be unaware of copyright violation?
- should we dispense with copyright as we have known it entirely and seek new paradigms?

Charging for information Charging for content is critical to the financial viability of many digital libraries as well as other (e.g. direct) providers of online content. Currently, there are a substantial number of different charging mechanisms available or proposed. Herzberg [88] investigates the need for charging mechanisms, in particular for micropayments (charging for small amounts), and describes two credit card charging mechanisms – using the SSL protocol (which is most common today) and the SET standard for charge card payments. He recommends the MiniPay micropayment mechanism for charging small amounts ‘per click’, and compares it with other micropayment mechanisms.

Authenticity/originality of information One of the major problems in a DL environment is to maintain or to ensure the originality of information. With the easy availability of technology, one can easily use information created by others and can claim it as one’s own. This has been an important concern, and some researchers have tried to address this problem. According to Mintzer et al. [89], digital watermarking can be used to indicate ownership and originality information. In the traditional manufacturing of paper, wet fibre is subjected to high pressure to expel the moisture. If the press’s mould has a slight pattern, this pattern leaves an imprint in the paper, a watermark, best viewed under transmitted light. Now, the old word ‘watermark’ has been borrowed by high technology. Digital

watermarks are imperceptible, or barely perceptible, transformations of digital data; often the digital data set is a digital multimedia object.

Gladney [90] reports on a suite of technical tools that address various perceived risks to the quality of digital information. Such a product, called 'document access control method' (DACM):

- mimics organisational practice by combining a subject tree with *ad hoc* role granting;
- controls privileges for many operations independently;
- treats privileged roles such as 'auditor' and 'security officer' like every other individual authorisation; and
- makes access control information part of ordinary objects.

However, the measures taken by publishers of incorporating some hidden data within resources and then using them for tracking the copyright violation may lead to the violation of privacy. Merry [91] argues that tracking of personal information attached to electronic text supplied by publishers is a potential violator of user privacy. Unmonitored in their efforts to protect their intellectual property, publishers could pose a threat to their customer's privacy and thus would limit their right of free access to information [91].

Social issues

As we build digital libraries, we need to be careful not to obliterate carelessly some of the important features of current libraries. Ackerman [92] suggests that many proposals for digital libraries remove social exchange and interaction, focusing narrowly on the technical mechanisms of information access, and he argues that this is not only unwise, but it is unnecessary since we could provide mechanisms for social exchange and interaction within our systems. Efthimiadis [93] discusses the social aspects of digital libraries and provides an overview of the developments in user-centred digital library research and the information life cycle model.

Ackerman [92] has argued that the design of a digital library does not have (and should not try) to eliminate the social world. Quite simply, there are important elements of the social world, including a sense of community, that we do not want to lose from our notions of 'library'. Many social mechanisms are important and useful in information access, and social interaction provides an enjoyable and community-building function. Ackerman [92] discusses a toolkit, called the Cafe ConstructionKit, that provides auxiliary mechanisms for seeking and locating information without adding significant programming or computational overhead. It is expected that the use of the social interaction capabilities of the Cafe ConstructionKit should make digital libraries more pleasant to use.

Evaluation of digital information

Traditionally, when scholars encountered original sources (artifacts, documents or works of art), many physical clues assisted in establishing their authenticity. If sources are studied in the surrogate (in photographs, microfilm or other reprographic form), questions concerning the authenticity of the original are overlaid

with questions about the methods of representation. With the proliferation of digital information sources on the Internet, these issues are further complicated [93]. Bearman and Trant [94] have raised a number of issues that have to be taken into consideration while ensuring the value of digital information. These issues, put in the form of questions, are: as creators of resources do we apply visible 'watermarks' as a deterrent against misuse; are we really looking for the potential evidentiary value of a hidden watermark or digital signature that can be cited in the case of a misuse or infringement; are we simply offering a 'seal of approval' indicating that a digital resource came from a trusted source; are we interested in a secure environment that guarantees the integrity of a retrieved resource, or are we more wedded to the philosophy of open access, and the flexibility that results from the ability to move a digital object between systems and platforms? All these boil down to the fact that a clear understanding of our requirements in particular circumstances is the only means to a comparative assessment of digital resources.

Evaluation of digital libraries

Digital libraries are quite different from more traditional (print) libraries and call for a new set of parameters for evaluation of their use. Lancaster [95] discusses evaluation criteria, problems and methods relevant to the digital library environment. Kelleher [96] summarises the evaluation of the eLib Programme, and describes the 'policy mapping' methodology of the review, the key findings and cultural issues and futures. House et al. [71] suggest that digital libraries can be described and evaluated on three key components: contents, functionality and interface. Usability assessment, as the term is used in the HCI community, usually addresses primarily interface design which is too narrow a basis for evaluating something as complex as a DL. Usability can be defined as a system's capability in human functional terms to be used easily and effectively by the specified range of users, given specified training and support, to fulfil a specified range of tasks, within the specified range of environmental scenarios. Evaluation of a DL's effectiveness has to be in terms of its impact on users' work. House et al. [71] recommend that for digital libraries to be truly useful, designers need first to understand the larger context that determines their information needs and purposes for using the DL, that is, the context of the users' work; the individual user's specific work and tasks; his or her information acts (including information searching, analysing, repackaging), and, finally his or her DL use.

Standards

Traditional libraries have developed and used several standards particularly with regard to the organisation of information resources, cataloguing, indexing, and so on. The need for standards and strict adherence to those is much more essential in a digital library environment. A large number of standards are involved in the different phases of a digital library, and librarians and users need to know them in order to avoid incompatibility and other problems. Dempsey et al. [97] provide recommendations for the selection and use of standards in eLib projects. They have identified and described several standards in the following areas of a digital library:

- application services:
 - basic applications;
 - search and request applications;
 - the World Wide Web;
- data interchange standards;
- metadata; and
- security, authentication and payment services.

Preservation

Some researchers consider digital libraries as an easy way of preserving information resources. For example, Hildreth [98] suggests that digital image formats are now seen not only as feasible alternatives to print and microfilm formats for traditional preservation purposes, but also as the technology bearing the potential for unlimited access and distribution. He further examines the requirements for browsing and exploration in the digital library of the future. Rutimann [99] introduces the history and work of the US Commission on Preservation and Access, established in 1986, and describes the work of its task force on archiving digital information, set up in 1994 to investigate means of ensuring continued access in the future to records stored in digital electronic form. The report attempts to answer three questions [99]:

- what does digital preservation entail;
- how archivists organise themselves to achieve it; and
- what infrastructure is needed.

However, there are some important issues of concern relating to the preservation of digital information resources. Lesk [23] argues that a digital library is more dynamic than a paper library: items will come and go more rapidly. Therefore we have to make a decision as to which item we shall preserve and how. The rapid developments of technology have a negative impact: technology becomes outdated too fast. We can read a book that was published, say three hundred years ago, but cannot read a text that was wordprocessed only five years ago, because the older versions of word processors are now obsolete. We used to keep back-up copies of our texts in 5.25" diskettes a few years ago, but they are no longer available now, and even if we have some such diskettes, we can hardly find a PC with that disk drive. This will continue to happen, probably faster, in future. Therefore, we have to be very careful in preserving digital information resources; and this looks to be a continuous problem.

Implications for library managers

The library and information service has always been considered as a personalised service. However, in a digital library environment, where end-users are expected to interact directly with computers for all of their information needs, the obvious question that comes up is: shall we have librarians, and if so what will be their roles? Matson and Bonski [100] discuss the effects of the new technology of digital libraries on librarians, and Sutton [101] discusses the role of reference librarians in digital libraries. Heckart [102] points out that as the infrastructure of a DL

takes shape, human help in user interactions will disappear and in future we shall see an almost total reliance on machine help. Griffin [103] comments that library managers are in the middle of a technology flux which can be compared to the invention and use of the printing press. He suggests that librarians in a digital library environment should: learn new vocabularies to understand and use the technology fully; encourage interaction among staff coming from the traditional library as well as technology background; carefully match technologies and users; reach out and educate users and staff; be prepared to lose control over materials that have so long been the library's property; and connect and communicate with the local, national and international professional events and work groups.

Ferguson and Bunge [104] argue that librarians must retain the timeless service values of equity of access, personal service and services tailored to the needs of individuals, and at the same time explore new values such as integrating technologies, maintaining holistic computing environments, delivering core services through the network, making technology work for all, and collaborating across administrative lines. Steele [105] suggests that in future the head librarian or information officer may well play the role of a chief executive officer (CEO) with the responsibilities and vision allied to more devolved structures. Steele further argues that knowledge managers will need to be constantly flexible and should realise the stated goals with an ever-broadening environment of delivery.

Davenport, Goldenberg and Procter [106] discuss the nature of reference work and the role of reference librarians in digital libraries where many users serve themselves by means of free-at-point-of-use information services. They consider how the concept of the 'reference desk' can be defined where points of presence for both users and librarians are distributed. They have reported the results of a study, undertaken in Edinburgh, Scotland, to explore these issues in the context of the enhanced regional communications available through EaStMAN (Edinburgh and Stirling Metropolitan Area Network). The project involved the BIOSIS Abstracts service hosted by the EDINA consortium and linked three university libraries (Edinburgh University, Heriot-Watt University and Napier University). The project investigated the experiences of users and the work patterns of librarians and related these to the design rationale of a prototype World Wide Web (WWW) based network reference consultation support system.

Smith [107] argues that the important interactions between librarians and users that occur in meta-information environments (an environment that comprises information about information, such as library catalogues, classification, and so on) in a traditional library may be lost with the automated information access in a digital library. It is therefore important to design special meta-information environments for digital libraries that can compensate for such loss. Walton and Edwards [108] and Hutingford [109] report a study, called IMPEL (Impact on People of Electronic Libraries), conducted in the UK as part of the eLib's supporting studies segment. IMPEL1 (1993–1995) focused on the changing role of qualified librarians in the higher education sector. IMPEL2 (1996–1997) (<http://www.unn.ac.uk/~lit8/impel2/impel2.html>) considered both library staff and users in the context of the changing library environment. The IMPEL study supports the proposition that the limiting factors for DL are human and organisational rather than technological.

It is evident that digital libraries will require new or specially trained manpower to develop and run them effectively and efficiently. Cronin [110] raises a question about the role of reference librarians and information intermediaries in a virtual world where collections are distributed on remote servers and intelligent agents are programmed to carry out searches for end-users and interactive recommender systems enable users to visualise and search social networks. As an answer to this, he suggests that some information professionals will be engaged in both the design and management of such software systems adding extra value through interpretation, evaluation and synthesis of automatically generated output.

Tennat [111] provides a list of skills, experience and qualification requirements of digital librarians. He further argues that it may be more productive if we choose staff who can evolve as the needs of the organisation change. Cronin [110] suggests that successful educational programmes geared to the needs of the digital age should be inclusive and tolerant of diverse perspectives, promoting interaction between disciplines and subfields in an effort to create discourse richness, experimentation and innovation.

Future directions

The discussion so far in this paper shows that DL research projects at government, institutional and individual levels throughout the world have led to the development of a wide range of digital library collections and services. Academic, special and public libraries are implementing online systems that provide their patrons with electronic access to library catalogues as well as to a variety of other information resources. Museums are digitising their collections and making them available on the Internet. Members of the various scientific communities are building collaborations to support their work and communication. Publishers are experimenting with the creation of digital archives of their journals and books, and individuals and groups from all walks of life are using community-based networks to provide local and global access to information resources they have created.

However, there are still a number of unresolved issues. Arms [112] argues that a great deal of effort in digital libraries is motivated by implicit assumptions about the market for information. It is also assumed that the creators and purveyors of high quality information are driven primarily by a quest for revenue. He argues that if we accept this view of information, then digital library research has to pay close attention to payment systems, security, encryption, authentication, licensing agreements and the resolution of legal questions of copyright and fair use.

Bishop [113] points out that digital libraries pose fascinating socio-technical challenges for understanding their use. The design and evaluation of digital libraries are complicated by the nature of the systems, their ability to integrate a range of functions that were previously designed and evaluated separately, the heterogeneity of their user population, the physically distributed nature of usage, the ability to fragment and rearrange previously integrated documents and images, and the rapid versioning of digital objects. Fortunately, appropriate user-centred research objectives, measures and methods for the digital library are just beginning to emerge. Results from user studies can help digital library designers and policy makers formulate appropriate goals, arrive at a more complete understanding of costs and benefits, design and allocate resources to both technologies

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and programmes that offer the best means of achieving goals, and assess the degree to which network policies and programmes have achieved their stated goals [113].

Now the question is how can we learn more about the use and users of digital libraries? How can people involved in user-centred studies of digital libraries share their ideas, concerns, methodological approaches and findings? These and many other questions can perhaps be answered once the results of the research projects in the second lap of DL initiatives, discussed below, and other parallel digital library research projects are completed and results made public.

Digital Libraries Initiative – phase 2 (DLI-2) The second phase of the DLI is jointly supported by the National Science Foundation (NSF), the Defense Advanced Research Projects Agency (DARPA), the National Library of Medicine (NLM), the Library of Congress (LC), the National Aeronautics and Space Administration (NASA), the National Endowment for the Humanities (NEH) and others (<http://www.nsf.gov/pubs/1998/nsf9863/nsf9863.htm>). Research areas in DLI-2 are organised into three areas for ease of exposition:

1. human-centred research that seeks to further understanding of the impacts and potential of digital libraries to enhance human activities in creating, seeking and using information and to promote technical research designed to achieve this;
2. content and collections-based research that focuses on better understanding of and advancing access to novel digital content and collections;
3. systems-centred research that focuses on component technologies and integration to realise information environments that are dynamic and flexible; responsive at the level of individual, group and institution; and capable of adapting large, amorphous, continually growing bodies of data to a user-defined structure and scale.

eLib phase 3 and the hybrid library concept Whereas phase 2 of eLib was designed to fill gaps in phase 1, phase 3 came somewhat later, with more of an aim to bring many separate developments together. Phase 3 of eLib was designed to have four major components [114]: hybrid libraries; large scale resource discovery, or clumps; preservation; and turning early projects into services. Rusbridge [17] mentions that the hybrid library effort, the attempts to scale digital library resources into production services, plus digital preservation, are the most important parts of the current eLib Programme. The term 'hybrid library' has been chosen to reflect the transitional state of the library, which today can neither be fully print nor fully digital. The challenge associated with the management of the hybrid library is to encourage end-user resource discovery and information use, in a variety of formats and from a number of local and remote sources, in a seamlessly integrated way [114]. The five hybrid library projects in the UK began in January 1998 and are due for completion in 2000. All these projects are facing the same broad issues such as authentication, user profiles, user interface design, database interconnectivity, digitisation management and staff development.

However, as Rusbridge [17] points out, each project concentrates on one specific area of research and development, such as:

- Agora (<http://hoste.ukoln.ac.uk/agora/>) is developing a hybrid library management system to provide integrated access to heterogeneous discovery, location, request and delivery services;
- BUILDER (Birmingham University Integrated Library Development and Electronic Resource) (<http://builder.bham.ac.uk/>) aims to build a working model of a hybrid library within both a teaching and research context;
- HeadLine (Hybrid Electronic Access and Delivery in the Library Networked Development) (<http://www.lse.ac.uk/blpes/headline.shtml>) aims to design and implement a working model of a hybrid library in actual academic environments, providing access to a wide range of library materials, regardless of physical form, through a single web interface;
- HyLife (Hybrid Library of the Future) (<http://www.unn.ac.uk/~xcu2/hylife/>) lays emphasis on the needs of users and on evaluation, at different levels, of the use and impact of the new services being developed;
- MALIBU (Managing the hybrid Library for the Benefit of Users) (<http://www.kcl.ac.uk/humanities/cch/malibu>) is focusing on the development of institutional models for the organisation and management of hybrid library services in humanities.

Pinfield et al. [114] discuss the development and various related issues of these five hybrid library projects. This paper shows that the issues facing these hybrid library projects are those that are universal to the digital library environment. For example, the user-centred approach and the emphasis on the management of organisational change is crucial to the development of digital libraries. Similarly, the technical area of authentication, addressed in the HeadLine and BUILDER projects is becoming an increasingly important issue. Interconnectivity and interoperability addressed in the BUILDER and Agora projects are essential to provide seamless access to the various digital information resources.

CONCLUSION

It is very difficult to draw any definite conclusion from an ongoing and fast developing research area like this. Significant amounts of funding and research efforts have been, and are continuing to be, pumped into digital library research resulting in proliferation of research publications on different issues of digital libraries. Cronin [110] points out that the digital library, in a sense, functions as a zone of convergence where librarians, computer scientists, electrical engineers, cognitive scientists, cultural anthropologists, organisational theorists and many others have come together for a better understanding of the nature of distributed information systems and knowledge access.

The present paper has provided just a snapshot of the ongoing research activities in this area, and has pointed to the various hot research issues. It can be noted that digital library research has drawn much attention in the most developed, and

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in a number of developing countries and government funding as well as institutional funding have come through. Initial digital library projects, such as the ELI-NOR project in the UK, the first two phases of the eLib Programme in the UK, and the first phase of DLI in the US are now over. Since the announcement of DLI, digital libraries research and application efforts have proliferated; new communities of researchers, information providers and users have become engaged; the definition of a digital library has evolved; technologies have advanced; stores of digital content have increased dramatically; and new research directions have emerged. However, much more research is needed before making digital libraries available and easily usable to everyone in society. In the US, such efforts are now being made through DLI-2 that aims to:

- provide leadership in research fundamental to the development of the next generation of digital libraries;
- advance the use and usability of globally distributed, networked information resources;
- encourage researchers to focus on innovative application areas to stimulate partnering arrangements necessary to create next-generation operational systems in such areas as education, engineering and design, earth and space sciences, bio-sciences, geography, economics, and the arts and humanities;
- address the digital libraries' life cycle from information creation, access and use, to archiving and preservation; and
- gain a better understanding of the long term social, behavioural and economic implications and effects of new digital libraries and their capabilities in such areas of human activities as research, education, commerce, defence, health services and recreation.

Rusbridge, the Director of the eLib Programme, has pointed out the following issues that need to be addressed and resolved through the digital library research in the UK [17]:

- quiet authentication providing authenticated access to information resources;
- rights or terms and conditions for metadata and automated licences;
- integrity and authenticity of electronic documents;
- useful copyright management systems;
- simple information trading including micro-payments;
- sophisticated electronic commerce;
- scalability issues, including collection summaries and query routing;
- metadata inter-operability;
- improved version of Z39.50;
- machine-independent, object-oriented packaging of digital library material; and
- digital preservation.

Taking all these into consideration, one can say that there are three major sets of issues that act as the stumbling blocks to the development, implementation and

successful use of digital libraries. One set of issues can be called the technical issues that call for the development of appropriate network, hardware, software, standards, tools and techniques related to the different aspects of digital libraries. These should be robust and yet quite user-friendly, and easy to adopt and use, for the generator as well as the user throughout the world. Library and information services have always been blessed with personalised services and support. Though this cannot be replaced by user interfaces, some of the problems that users face can be overcome with the development of sophisticated user interfaces, help and tutorial systems and intelligent agents; but this would require more research involving not only system developers but also the users across the board.

The second set of issues raises a number of questions that relate to the conversion of traditional libraries into digital libraries, such as:

- shall we leave the present printed libraries as they are and build digital libraries alongside so that users will have to use both of them to get the required information;
- shall we build digital libraries only in certain specific areas and let the traditional libraries develop at their own pace; or
- shall we convert the existing libraries to digital libraries so that everything will be digital?

All these take us to the third set of issues that is concerned with the economic, social and legal aspects of digital libraries. On the economic front, the questions are how much can we spend, and how much shall we spend, to develop digital libraries? It may be noted that several million dollars have been spent so far just to build prototypes and to develop our understanding of certain issues related to digital libraries. Can we afford to continue to do so, particularly when libraries have continually been hit by budget cuts throughout the globe (unfortunately libraries are the first target for cost reduction measures, and this is more so when the whole world is facing recession of some sort or the other). The other, and perhaps more difficult economic question, relates to the need for changes in the multi-billion dollar publication industry and the related business that is necessary if we want to go digital leaving behind the conventional printed world. Are we ready for that, and can we control the payment mechanisms, profit, loss, etc., with the existing regulatory and management mechanisms, or do we need to bring in new mechanisms and management techniques? The question of authenticity and reliability of information will also become critical in digital libraries, and Lesk [23] has predicted that the job of evaluating and assessing information will become more important to everyone.

Griffin [103] comments that the social and economic impact of digital library research is likely to be unprecedented because of the scale involved. Perhaps the most critical issue relates to the social order and the need for change of habits and human behaviour. Human society has been accustomed to hard copies of documents for centuries. How quickly can we change our generation-old habits and switch to a new world of digital and only digital information resources? As Pinfield et al. [114] have pointed out, users themselves need to be assisted to develop new approaches to seeking and using information sources.

To sum up, it can be stated that digital libraries are still very much at the experimental level, and we still have to wait for some more time before we can see fully-implemented digital libraries. Then will come the period when we have to face some teething problems, and simultaneously we have to change our generation-long habits and have to get used to our new shoes. This will take some time, because it will involve a paradigm shift in our habits of the creation, distribution and use of information. In the meantime, digital library researchers – librarians, information professionals, computer scientists, engineers, lawyers, educators, publishers and the like – have to work hand-in-hand with the most important component of digital libraries, the end-users, to make things happen fast and to the best of human civilization.

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